

Analytic Functions of a Complex Variable

Prof. V. Balakrishnan

Handwritten lecture notes, transcribed

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(Analytic functions of a complex variable)

Note – (Mod 01 – lec 1) of this topic is written in lec 25 of Quantum mechanics.

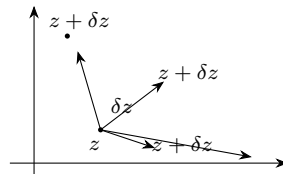
1 Module 1, Lecture 2

outline :-

- (1) The derivative of an analytic function
- (2) power series as analytic functions
- (3) convergence of power series.

(1) Derivative of analytic function :-

let $f(z)$ be analytic function in complex plane $\bar{z}$.



$$\frac{df}{dz} = \lim_{\delta z \rightarrow 0} \frac{f(z + \delta z) - f(z)}{\delta z}$$

Here we can see that we do not know in which direction we should proceed to calculate derivative?

let us take a general case

$$\delta z = \epsilon e^{i\alpha}, \quad z = x + iy$$

$$\begin{aligned} \frac{df}{dz} &= \lim_{\delta z \rightarrow 0} \frac{f(z + \delta z) - f(z)}{\delta z} = \\ &= e^{-i\alpha} \left\{ \left(\frac{\partial u}{\partial x} \cos \alpha + i \frac{\partial v}{\partial y} \sin \alpha \right) - i \left(\frac{\partial u}{\partial y} \cos \alpha - i \frac{\partial v}{\partial x} \sin \alpha \right) \right\} \end{aligned}$$

What direction should α be.

One way to define the derivative as it is same in all directions specified by ' α ', i.e. it is independent of α .

$$\frac{df}{dz} = \text{independent of } \alpha \quad \text{when} \quad \left. \begin{array}{l} u_x = v_y \\ u_y = -v_x \end{array} \right\} \underline{\text{CR Relations}}$$

i.e. For analytic function their derivative does not depend on direction specified by angle α .

Some Real functions are not infinitely differentiable.

Ex $f(x) = x^2$ (only twice differentiable)

$$f'(x) = 2x$$

$$f''(x) = 2$$

$$f'''(x) = 0$$

$\therefore$ and so does $f^n(x)$ does not exist.

In complex functions we do not have such complications.

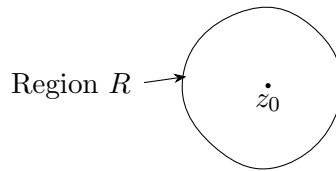
If $f(z)$ is analytic $\Rightarrow f'(z)$ is analytic $\Rightarrow f''(z)$ is analytic

The derivative of analytic function is also analytic function so does $f''(z)$ & higher derivatives.

so $f(z)$ is analytic ($\Leftrightarrow$) all its derivative are also analytic.

Power series representation :-

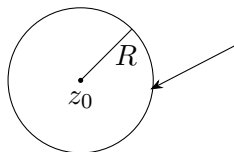
We can always represent a given analytic function in some power series about some point z_0 which lies inside some region of analyticity / region of convergence.



$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n \quad (\text{Taylor series})$$

$$\left(a_n = \frac{1}{n!} \left. \frac{\partial^n f(z)}{\partial z^n} \right|_{z=z_0} \right) \quad (\text{Inverse of Taylor series})$$

But this power series should be convergent to some finite value within some region of analyticity / circle of convergence about point z_0 with radius R . such that



$$\left(\sum_{n=0}^{\infty} |a_n (z - z_0)^n| < \infty \right)$$

To find radius of convergence $|z - z_0|$ we need,

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1} (z - z_0)^{n+1}}{a_n (z - z_0)^n} \right| < 1$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| |z - z_0| < 1$$

$$\left(R < \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| \right)$$

↓

radius of convergence.

Now there is no guarantee that this $\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|$ exists

So we define

$$\left(R = \limsup_{n \rightarrow \infty} |a_n|^{1/n} \right)$$

If radius of convergence for a given $f(z)$ is ' ∞ ' then this function is entire function (defined at ∞ too).

Examples:

(1) $f(z) = e^z$

$$e^z = \sum_n \frac{z^n}{n!}$$

to converge

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{z^{n+1}}{(n+1)!} \left(\frac{n!}{z^n} \right) \right| &< 1 \\ \Rightarrow \lim_{n \rightarrow \infty} \left| \frac{z}{n+1} \right| &< 1 \end{aligned}$$

for all finite value of z , ($z < \infty$)

$$\lim_{n \rightarrow \infty} \left| \frac{z}{n+1} \right| \rightarrow 0 < 1 \quad (\text{converges}).$$

(Radius of convergence) :-

$$\begin{aligned} R &\leq \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| \\ R &\leq \lim_{n \rightarrow \infty} \left| \frac{z^n (n+1)!}{n!} \right| \\ (R \rightarrow \infty) &\quad (\text{entire function}). \end{aligned}$$

(2) $f(z) = 1 + z + z^2 + z^3 + \dots \infty = \sum_{n=0}^{\infty} z^n$

Is this function entire function?

Is the series convergent, where does this series converge?

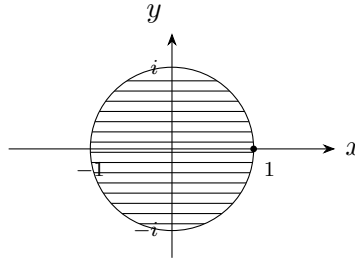
$f(z)$ converges for $|z| < 1$.

we can see

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{z^{n+1}}{z^n} \right| &< 1 \\ \left(\lim_{n \rightarrow \infty} |z| < 1 \right) \end{aligned}$$

So Radius of convergence should be $R = 1$.

check, $R \leq \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| \Rightarrow (R = 1) \quad (\text{as } a_n = a_{n+1} = 1).$



Inside this region series $\sum_{n=0}^{\infty} z^n$ is convergent.

$$f(z) = \sum_{n=0}^{\infty} z^n \quad \text{diverges for all } z \text{ such that } \underline{|z| > 1}$$

Inside region of convergence / analyticity :

we can identify that

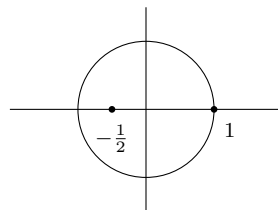
$$f(z) = 1 + z + z^2 + \dots = \frac{1}{1-z}$$

$$f(z) = \begin{cases} \sum_{n=0}^{\infty} z^n & (|z| < 1) \\ \frac{1}{1-z} & (z \neq 1) \text{ \& } (|z| < 1) \end{cases}$$

so the representation of $f(z) = \frac{1}{1-z}$ matches with $f(z) = \sum_{n=0}^{\infty} z^n$ point by point within region $(|z| < 1)$ or $\underline{|(z-0)| < 1}$.

But $g(z) = \frac{1}{1-z}$ is making sense / definable at all z . except $\underline{z=1}$. So, $g(z) = \frac{1}{1-z}$ is called as analytic continuation of $f(z)$.

writing power series $f(z) = 1 + z + z^2 + \dots$ about $(z_0 = -\frac{1}{2})$.



within region $|z| < 1$

$$\begin{aligned} f(z) &= \frac{1}{1-z} \\ &= \frac{1}{1 - (z - (-\frac{1}{2})) + \frac{1}{2}} \\ f(z) &= \frac{1}{1-z} = \frac{1}{1 + \frac{1}{2} - \frac{1}{2} - z} = \frac{1}{\frac{3}{2} - (z + \frac{1}{2})} = \frac{1}{\frac{3}{2} (1 - \frac{2}{3} (z + \frac{1}{2}))} \\ f(z) &= \frac{2}{3} \left(1 - \frac{2}{3} \left(z + \frac{1}{2} \right) \right)^{-1} \end{aligned}$$

$$((1-x)^{-1} = 1 + x + x^2 + \dots)$$

$$f(z) = \frac{2}{3} \sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n \left(z + \frac{1}{2}\right)^n$$

$\hookrightarrow$ power series about $(z_0 = -\frac{1}{2})$

where does this series converge.?

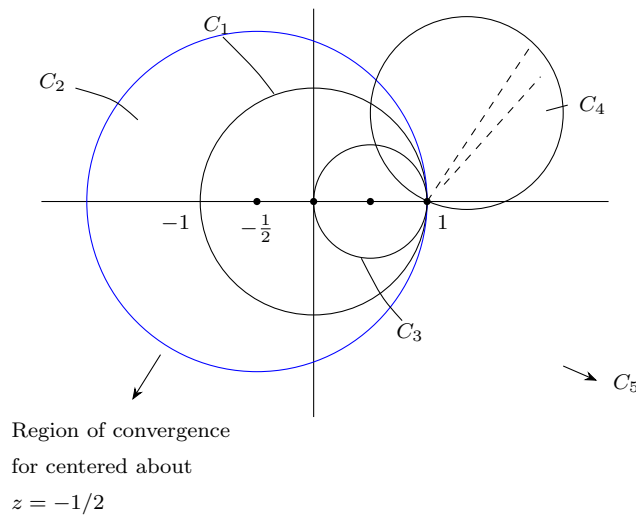
for convergence

$$\lim_{n \rightarrow \infty} \left| \frac{\left(z + \frac{1}{2}\right)^{n+1} \left(\frac{2}{3}\right)^{n+1}}{\left(z + \frac{1}{2}\right)^n \left(\frac{2}{3}\right)^n} \right| < 1$$

$$\lim_{n \rightarrow \infty} \left| \left(z + \frac{1}{2}\right) \left(\frac{2}{3}\right) \right| < 1$$

$$\left(\left| z + \frac{1}{2} \right| < \frac{3}{2} \right)$$

$\hookrightarrow$ circle of radius $3/2$ centered at $(z = -1/2)$



We can write the power series

$$f(z) = 1 + z + z^2 + \dots \infty$$

		Region of convergence / circle of converg[?]	
centered about	$z = 0$	$\longrightarrow$	$C_1 : z < 1$
	$z = -1/2$	$\longrightarrow$	$C_2 : \left z + \frac{1}{2} \right < \frac{3}{2}$
	$z = 1/2$	$\longrightarrow$	$C_3 : \left z - \frac{1}{2} \right < \frac{1}{2}$
	$z = 3/2 + i$	$\longrightarrow$	$C_4 : \left z - \left(\frac{3}{2} + i\right) \right < \frac{\sqrt{5}}{2}$
	$z = 2 - i$	$\longrightarrow$	$C_5 : z - (2 - i) < \sqrt{5}$

All these representations are valid for given power series but they are valid in given Region of analyticity or convergence given by region C_i , $1 \leq i \leq 5$.

Note :: So A given $f(z)$ can have infinite no. of representations. valid in different regions.

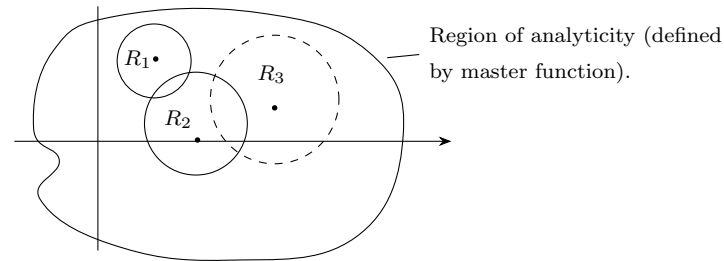
All these regions / representations are analytic continuations of each other.

We may or may not be able to find master representations from which you can find other representations. For $f(z) = 1 + z + z^2 + \dots$ (master representation is $f(z) = \frac{1}{1-z}$). A given representation of function is valid in some region of convergence so power series diverges outside region of convergence & converges for all 'z' inside circle of convergence. We are not sure about behaviour of $f(z)$ at boundary of convergence.

Region of convergence $\in$ Region of analyticity.

Region of convergence $<$ Region of analyticity.

$f(z) = 1 + z + z^2 + \dots \infty$ is only valid within region ($|z| < 1$).



R_1 & R_2 , R_3 are regions of convergence of given power series about some different z_{01}, z_{02}, z_{03} .

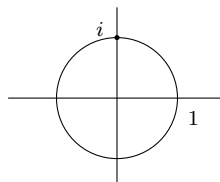
Behaviour at Boundary :-

(1) On boundary of circle of convergence., The series provides some use. (If series oscillates).

for, $f(z) = 1 + z + z^2 + z^3 + \dots$

$$f(z) = \frac{1}{1-z} \quad (\text{master representation})$$

at ($z = 1$) $f(z) \rightarrow \infty$ $\left(\begin{array}{c} \text{singularity} \\ \text{at } z = 1 \end{array} \right)$



(a) $f(-1) = 1 - 1 + 1 - 1 + 1 - 1 + \dots$ $\checkmark$ Cisaró sum

lets take average of outcomes.

Possible outcomes = 1, 0

$$\text{avg} = \frac{1+0}{2} = 1/2$$

$$f(-1) = 1/2$$

we can check via $f(z) = \frac{1}{1-z} \Big|_{z=-1} = \frac{1}{2}$

(b) $f(i) = 1 + i + i^2 + i^3 + i^4 + \dots$ $\swarrow$ Cisaró sum.

$$= 1 + i - 1 - i + \underline{1} + \dots$$

Possible outcomes = 1, $1 + i$, i , 0,

$$f(i) = \text{avg}(1, 1 + i, i, 0) = \frac{2(1 + i)}{4} = \frac{i + 1}{2}$$

$$f(i) = \frac{1}{1 - i} = \frac{1 + i}{1 - i^2} = \frac{1 + i}{2}$$

Theorem : Any power series representation of $f(z)$ must have at least one singularity on circle of convergence.

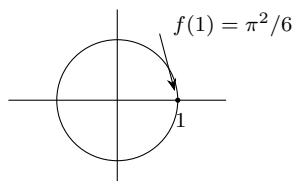
(2) Convergence / absolute convergence is also possible for z which lies on boundary of circle of convergence.

Ex $f(z) = \sum_{n=0}^{\infty} \frac{z^n}{n^2}$ converges for $\left| \frac{z^{n+1}}{(n+1)^2} \frac{n^2}{z^n} \right| < 1$

$$\lim_{n \rightarrow \infty} \left| \frac{z n^2}{(n+1)^2} \right| < 1$$

converges for. $|z| < 1$

In fact it converges even for points $|z| = 1$. too.



But $\exists$ a point on boundary $|z| = 1$ such that It has a singularity.

(In fact $f(z)$ is singular at $z = 1$.)

If we have $g(z) = (z - 1) \ln(z - 1)$

$$\left(\lim_{z \rightarrow \underline{1}} g(z) = 0 \right)$$

as $(z \rightarrow \underline{1}) \quad g(z) \rightarrow 0$

as log is weaker than power

$\log(z - 1)$ is weaker than $(z - 1)$.

Singular part is zero. But it is singular at $\underline{z = 1}$

Singularity does not necessarily mean $f(z) \rightarrow \infty$, it is singular in sense of theory of analytic functions.

(Ex 2) $f(z) = z + z^2 + z^4 + z^8 + \dots$

$$f(z) = \sum_{n=0}^{\infty} z^{2^n} < \infty \text{ for } \left| \frac{z^{2^{n+1}}}{z^{2^n}} \right| < 1$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left| \frac{z^{2^n} \cdot z^{2^n}}{z^{2^n}} \right| < 1$$

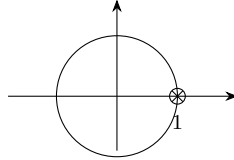
(Radius of convergence = 1)

$$\lim_{n \rightarrow \infty} |z^{2^n}| < 1$$

only if $|z| < 1$

for $(z = 1)$ $f(z) = 1 + 1 + 1 \cdots$ diverges.

($f(z)$ is singular at $z = 1$)



we can also write

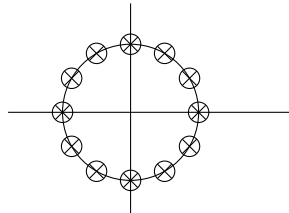
$$f(z) = z + f(z^2)$$

$$f(z^2) = z + z^2 + z^3 + \cdots$$

In this, we have singularity at $z = \pm 1$

Similarly $f(z) = z + z^2 + f(z^4)$

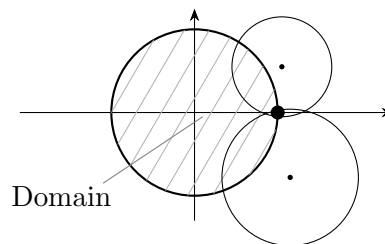
($\rightarrow$ have more singular points.



In this way $|z| = 1$ forms natural boundary where it is singular at infinite points on boundary $|z| = 1$.

So for $f(z) = z + z^2 + z^4 + z^8 + \cdots$

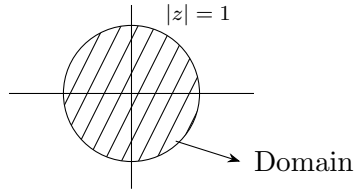
we can't have any other representation except as shown in below fig. for $f(z)$.



There is no analytic function continuation possible beyond the natural boundary due to presence of infinite singular points on $|z| = 1$.

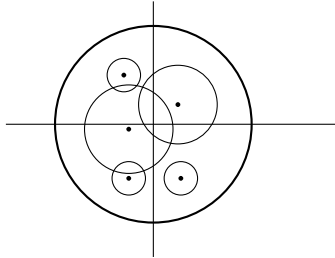
$$f(z) = \sum_{n=0}^{\infty} z^{n!} \quad \text{converges for} \quad \left| \frac{z^{(n+1)!}}{z^{n!}} \right| < 1$$

$$|z| < 1$$



for this $f(z)$ too we have infinite no of singular points on $|z| = 1$, so we only have this one representation of $f(z)$.

so $\exists$ such $f(z)$ for which we can only have one power series expansion.

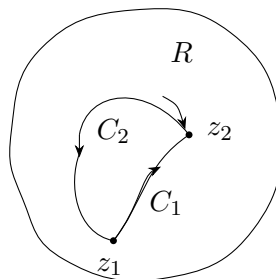


Inside we can have many representations.

2 Module 1, Lecture 3: Calculus of Residues

We have seen that if a function $f(z)$ is analytic in some region, then C-R conditions are satisfied, derivative is uniquely defined at every point, derivative itself is an analytic function & $f(z)$ is infinitely differentiable.

Integral of analytic function $f(z)$:-



$\left(\int_{z_1}^{z_2} f(z) dz \right)$ is independent of path 'C'.

It is similar to line integral of $\nabla\phi(r, t)$.

$$\int_a^b \nabla\phi \cdot d\vec{r} = \phi(b) - \phi(a)$$

(Cauchy's Integral theorem):

So $I = \oint f(z) dz = 0$ (Cauchy's Integral theorem)

$$\begin{aligned} \text{as } I &= \int_{z_1(C_1)}^{z_2} f(z) dz + \int_{z_2(C_2)}^{z_1} f(z) dz \\ &= \int_{z_1(C_1)}^{z_2} f(z) dz - \int_{z_1(C_2)}^{z_2} f(z) dz \\ &= 0 \quad \text{as} \quad \int_{z_1(C_1)}^{z_2} f(z) dz = \int_{z_1(C_2)}^{z_2} f(z) dz. \end{aligned}$$

we can actually deform the path / contour of integration without changing its value as long as path remains in analytic region.

Singularity :

(A) Removable singularities :- upon proper redefinition of function this singularity can be removed.

example :

$$(1) \quad f(z) = \frac{\sin z}{z}$$

at $z = 0$ $f(z)$ is having removable singularity

$$\text{as } \lim_{z \rightarrow 0} f(z) = \lim_{z \rightarrow 0} \frac{\sin z}{z} = \lim_{z \rightarrow 0} \frac{\cos z}{1} = 1$$

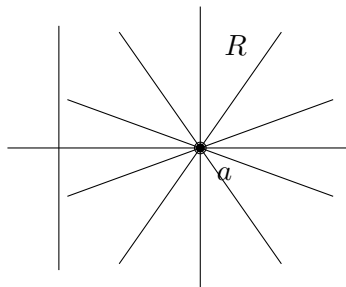
$$\text{If we redefine, } f(z) = \begin{cases} \frac{\sin z}{z} & z \neq 0 \\ 1 & z = 0 \end{cases}$$

this function is now free from singularity.

(B) Simple pole :-

If a function $f(z)$ is not analytic at $z = a$ and analytic everywhere / or around it for some radius of convergence R . then

(R may go to ∞).



Then we can express such $f(z)$

$$f(z) = \underbrace{\frac{C_{-1}}{(z-a)}}_{\text{singular part}} + \underbrace{\sum_{n=0}^{\infty} C_n (z-a)^n}_{\text{Regular part}}$$

$f(z)$ is said to have simple pole at $\underline{z=a}$.

$$f(z) = \frac{\boxed{c_{-1}}}{(z-a)^{(1)}} + \sum_{n=0}^{\infty} c_n (z-a)^n$$

→ Residue (pointing to c_{-1}); ↓ Simple pole at $z=a$ (pointing to $(z-a)^{(1)}$)

and, residue at the pole $z=a$ is c_{-1}

Singularity may not be always obvious.

example

$$(1) f(z) = \frac{\sin z}{z^2}$$

$$= \frac{\left(z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots \right)}{z^2}$$

$$= \left(\frac{1}{z} \right) + \left\{ -\frac{z}{3!} + \frac{z^3}{5!} - \dots \right\}$$

↓ Singular part (pointing to $\frac{1}{z}$); ↪ Regular part (under the braced terms)

Simple pole at $z=0$

with residue = $\underline{1}$

$$(2) f(z) = \frac{g(z)}{h(z)}$$

lets say, $g(a) \neq 0$ & $h(a) = 0$

Then,

$$f(z) = \frac{g(a) + (z-a)g'(a) + \frac{(z-a)^2}{2!}g''(a) + \dots}{\overset{0}{h(a) + (z-a)h'(a) + \dots}}$$

$$= \lim_{z \rightarrow a} \frac{g(a) + (z-a)g'(a) + \dots}{(z-a)h'(a) + \dots}$$

as $\lim z \rightarrow a$

$$g(z) \simeq g(a)$$

$$h(z) \simeq (z-a)h'(a)$$

$$f(z) \simeq \frac{g(a)}{h'(a)} \frac{1}{(z-a)}$$

→ simple pole at $\underline{z=a}$

with residue $\frac{g(a)}{h'(a)}$

$$\left\{ \begin{array}{l} \text{In general, for Simple pole at } z = a \\ \text{The residue} = \lim_{z \rightarrow a} (z - a) f(z) \end{array} \right\}$$

as

$$f(z) = \frac{c_{-1}}{(z-a)} + \sum_{n=0}^{\infty} c_n (z-a)^n$$

$$(z-a)f(z) = c_{-1} + \sum_{n=0}^{\infty} c_n (z-a)^{n+1}$$

$$\lim_{z \rightarrow a} (z-a)f(z) = c_{-1} = \text{Residue at simple pole.}$$

(3) $f(z) = \frac{1}{\sin \pi z}$ Find z for which $f(z)$ is singular and find Residue at each pole.

$f(z)$ is singular for $\pi z = (\text{Integer}) \pi$

or $z = \text{integer}$ (all integers)

or $z = n, \quad n = \text{Integer}$

$f(z)$ has simple poles at all integers.

$$\text{as } z \rightarrow 0, \sin \pi z \rightarrow (\pi z) \quad \text{as } \left(\sin \pi z = \pi z - \frac{(\pi z)^3}{3!} + \dots \right)$$

$$\text{so as } z \rightarrow 0, f(z) \cong \frac{1}{\pi z}$$

so simple pole at $(z=0)$ with residue $\left(\frac{1}{\pi}\right)$.

But to find Residue at all simple poles at $z = n$ where $n \in \text{Integers}$, $(\dots, -3, -2, -1, 0, 1, 2, 3, \dots)$.

$$\begin{aligned} \text{Res}(z=n) &= \lim_{z \rightarrow n} (z-n)f(z) \\ &= \lim_{z \rightarrow n} (z-n) \cdot \frac{1}{\sin \pi z} \\ &= \lim_{z \rightarrow n} \frac{1}{\pi \cos \pi z} = \frac{(-1)^n}{\pi} \\ &\left(\text{Res}(z=n) = \frac{(-1)^n}{\pi} \right). \end{aligned}$$

(C) Multiple poles

$$f(z) = \frac{c_{-m}}{(z-a)^m} + \dots + \frac{\boxed{c_{-1}}}{(z-a)} + \underbrace{\sum_{n=0}^{\infty} c_n (z-a)^n}_{\text{Regular part.}}$$

✓ Residue (pointing to c_{-1}); ↓ ($m > 0$) (pointing to $(z-a)^m$)

$f(z)$ is said to have pole of order m at $z = a$.

But Residue is only defined for *pole of order 1 or simple pole*

$$c_{-1} = \text{Res}(z=a) = \lim_{z \rightarrow a} \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} [(z-a)^m f(z)]$$

(D) Essential / Isolated singularity :-

Going a step further, the singular part may involve all negative Integral powers of $(z-a)$.

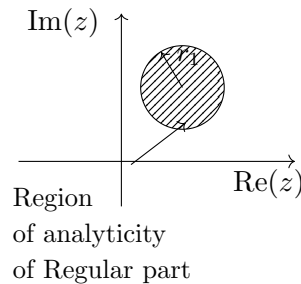
The function $f(z)$ has an isolated/essential singularity at $z=a$ if, in the neighbourhood of that point it can be expressed in form

$$f(z) = \underbrace{\sum_{n=1}^{\infty} \frac{c_{-n}}{(z-a)^n}}_{\text{Singular part.}} + \underbrace{\sum_{n=0}^{\infty} c_n (z-a)^n}_{\text{Regular part}} \quad \left\{ \begin{array}{l} \text{Laurent} \\ \text{series} \end{array} \right\}$$

regular part $\sum_{n=0}^{\infty} c_n (z-a)^n$ converges for $\left| \frac{c_{n+1}(z-a)^{n+1}}{c_n(z-a)^n} \right| < 1$

$\forall z$ which lies inside region

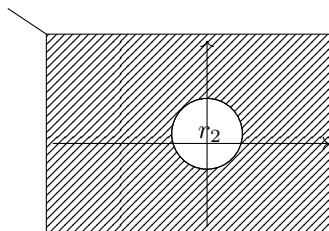
$$\begin{aligned} |z-a| &< \lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right| & |z-a| &< \left| \frac{c_n}{c_{n+1}} \right| \\ &(|z-a| < r_1) & \text{or} & |z-a| < r_1 \end{aligned}$$



But the singular part, $\sum_{n=1}^{\infty} \frac{c_{-n}}{(z-a)^n}$ converges for

$$\lim_{n \rightarrow \infty} \left| \frac{c_{-n-1} (z-a)^n}{(z-a)^{n+1} c_{-n}} \right| < 1$$

Region of convergence
for singular
part



$$\lim_{n \rightarrow \infty} \left| \frac{1}{(z-a)} \right| \left| \frac{c_{-n-1}}{c_{-n}} \right| < 1$$

$$\text{or } |z-a| > \lim_{n \rightarrow \infty} \left| \frac{c_{-n-1}}{c_{-n}} \right|$$

$$(|z-a| > r_2)$$

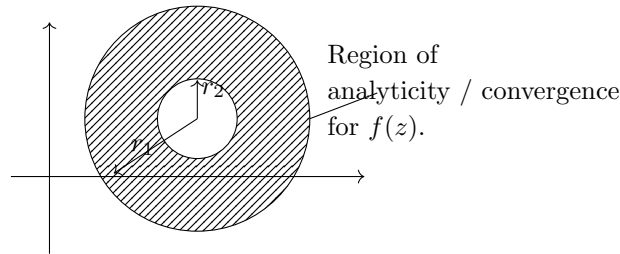
converges for,

To have representation of $f(z) = \sum_{n=1}^{\infty} \frac{c_{-n}}{(z-a)^n} + \sum_{n=0}^{\infty} c_n (z-a)^n$

in this laurent series, we need both regions to overlap.

To overlap ($r_2 < r_1$),

$$\lim_{n \rightarrow \infty} \left| \frac{c_{-n-1}}{c_{-n}} \right| < \lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right|$$



$$f(z) = \sum_{n=1}^{\infty} \frac{c_{-n}}{(z-a)^n} + \sum_{n=0}^{\infty} c_n (z-a)^n$$

So region of convergence becomes, ($r_2 < |z-a| < r_1$)

It is possible that $r_2 \rightarrow 0$ & $r_1 \rightarrow \infty$.

For example

(1) $f(z) = e^{\frac{1}{z}}$

$$f(z) = \underbrace{1}_{\text{Regular part}} + \underbrace{\frac{1}{z} + \frac{1}{2!} \left(\frac{1}{z}\right)^2 + \frac{1}{3!} \left(\frac{1}{z}\right)^3 + \dots}_{\text{(Singular part)}}$$

↓ ($\cdot$ Regular part) (pointing to the 1)

$$= \underbrace{\sum_{n=1}^{\infty} \frac{1}{n!} z^n}_{\text{Singular part}} + \underbrace{\boxed{1}}_{\text{Regular part}} \longrightarrow \text{converges everywhere}$$

↓

converges for

$$\lim_{n \rightarrow \infty} \left| \frac{(n)! z^n}{(n+1)! z^{n+1}} \right| < 1$$

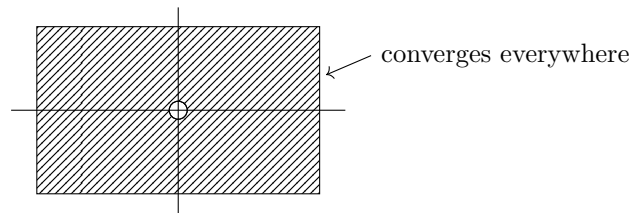
$$\lim_{n \rightarrow \infty} \left| \frac{1}{(n+1)z} \right| < 1$$

$$|z| > \lim_{n \rightarrow \infty} \left| \frac{1}{n+1} \right|$$

$$(|z| > 0) \quad \text{i.e. all } z \text{ except } \underline{(z \neq 0)} \quad z = 0.$$

$f(z)$ converges $\forall z \neq 0$. including point at infinity.

So Region of analyticity/convergence is a punctured disk.



Q) What is the residue ?

Residue is coefficient of $\frac{1}{z}$ which is 1 for $f(z) = e^{1/z}$.

$$(2) f(z) = e^{1/z} + e^z$$

$$= 1 + \sum_{n=1}^{\infty} \frac{1}{n!} \frac{1}{z^n} + \left(1 + z + \frac{z^2}{2!} + \dots \right)$$

$$= \underbrace{\sum_{n=1}^{\infty} \frac{1}{n!} \frac{1}{z^n}}_{\text{Singular part}} + \underbrace{2 + \sum_{n=1}^{\infty} \frac{z^n}{n!}}_{\text{Regular part}}$$

↓ Converges for (singular part)

$$\lim_{n \rightarrow \infty} \left| \frac{1}{(n+1)!} \frac{1}{z^{n+1}} \cdot n! z^n \right| < 1$$

$$\lim_{n \rightarrow \infty} \left| \frac{1}{(n+1)z} \right| < 1$$

$$\text{or } |z| > \lim_{n \rightarrow \infty} \left| \frac{1}{n+1} \right|$$

$$(|z| > 0) \quad \text{i.e. } (\forall z \neq 0)$$

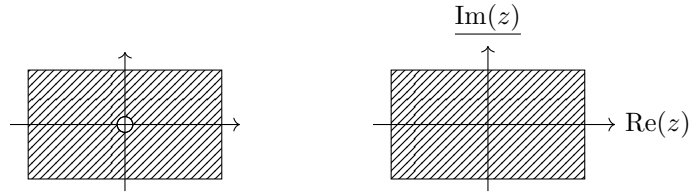
↓ Converges for (regular part)

$$\lim_{n \rightarrow \infty} \left| \frac{z^{n+1}}{(n+1)!} \frac{n!}{z^n} \right| < 1$$

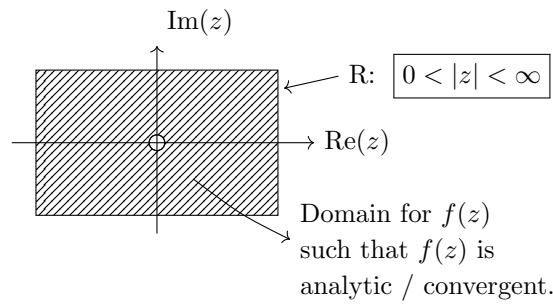
$$\lim_{n \rightarrow \infty} \left| \frac{z}{n+1} \right| < 1$$

$$|z| < \lim_{n \rightarrow \infty} |n+1|$$

$$(|z| < \infty) \text{ i.e. } \forall |z| < \infty.$$



So $f(z) = e^{1/z} + e^z$ converges for the common region



So on Riemann sphere $z = 0$ is south pole & $z = \infty$ is north pole

So $f(z)$ converges $\forall z$ except North pole & south pole.

$$(3) f(z) = e^{\frac{1}{z^2}}$$

$$= 1 + \frac{1}{z^2} + \frac{1}{2!} \frac{1}{z^4} + \frac{1}{3!} \frac{1}{z^6} + \dots$$

$$= \sum_{n=1}^{\infty} \frac{1}{n!} \frac{1}{z^{2n}} + \underbrace{1}$$

↓ Converges $\forall z$ including ∞ . (under the 1)

↓

Converges for $|z| > 0$

→ $f(z)$ has essential singularity (non removable). as its singular part is summed over $\left(\sum_{n=1}^{\infty} \frac{1}{z^{2n}} \frac{1}{n!} \right)$.

→ Residue = coeff of $\frac{1}{z} = 0$.

$f(z)$ converges for $(0 < |z| < \infty)$

$$\begin{aligned}
(4) \quad f(z) &= e^{(-)\frac{1}{(z-2)^2}} \\
&= 1 - \frac{1}{(z-2)^2} + \left(\frac{1}{2!}\right) \cdot \frac{1}{(z-2)^4} - \frac{1}{3!} \left(\frac{1}{(z-2)^6}\right) + \dots \\
&= \underbrace{\sum_{n=1}^{\infty} \frac{(-1)^n}{(z-2)^{2n}} \left(\frac{1}{n!}\right)}_{\text{Singular part}} + \underbrace{1}_{\text{Regular part}}
\end{aligned}$$

So $f(z)$ has essential singularity at $z = 2$

& Residue = coeff of $\frac{1}{(z-2)} = 0$

$$\begin{aligned}
(5) \quad f(z) &= (z-2) e^{\frac{-1}{(z-2)^2}} \\
&= (z-2) \left(\sum_{n=1}^{\infty} \frac{(-1)^n}{(z-2)^{2n}} + 1 \right) \\
&= (z-2) \left(1 - \frac{1}{(z-2)^2} + \frac{1}{2! (z-2)^4} \dots \right) + (z-2) \\
&= \underbrace{\left(\frac{-1}{(z-2)} + \frac{1}{2! (z-2)^3} - \dots \right)}_{\text{Singular part}} + \underbrace{(2z-4)}_{\text{Regular part.}}
\end{aligned}$$

It has essential singularity at $z = 2$

Residue = coeff of $\left(\frac{1}{z-2}\right) = -1$

$$(6) \quad f(z) = (z-2)^4 e^{\frac{-1}{(z-2)^2}}$$

Again $f(z)$ involves infinite sum for singular part

So it has essential singularity.

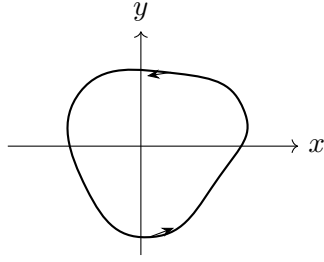
$$\begin{aligned}
f(z) &= (z-2)^4 \left[1 - \frac{1}{(z-2)^2} + \frac{1}{2!} \left(\frac{1}{(z-2)^4} \right) - \frac{1}{3! (z-2)^6} + \dots \right] \\
&= \underbrace{\left((z-2)^4 - (z-2)^2 + \frac{1}{2} \right)}_{\text{Regular part}} - \underbrace{\frac{1}{3! (z-2)^2} + \dots}_{\text{Singular part.}}
\end{aligned}$$

Residue = coeff of $\left(\frac{1}{z-2}\right) = 0$.

Cauchy's Integral formula :-

Consider integral, $\oint z^n dz$

$$I = \left(\oint z^n dz \right) =$$

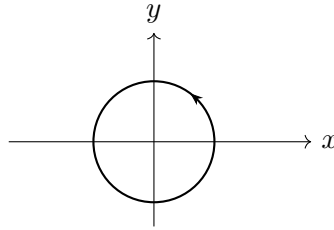


$$(n = 0, 1, 2, 3 \dots)$$

Since z^n is analytic everywhere (entire function)

$$\oint z^n dz = 0$$

we can distort the contour and shrink it to a point, or we can make it a circle $|z| = 1$.



On this unit circle $z = e^{i\theta}$ ($r = 1$)

$$dz = e^{i\theta} i d\theta$$

$$I = \oint z^n dz = \int_0^{2\pi} e^{(n+1)i\theta} i d\theta = i \int_0^{2\pi} (\cos(n+1)\theta + i \sin(n+1)\theta) d\theta$$

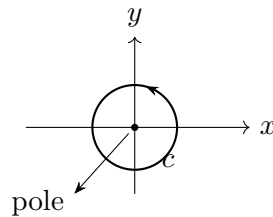
$$I = i \left(\frac{\sin(n+1)\theta}{n+1} + \frac{i \cos(n+1)\theta}{n+1} \right)_0^{2\pi} = 0$$

what happens if

$$I = \oint \frac{dz}{z^{n+1}} \quad (n = 0, 1, 2, 3 \dots)$$

$f(z) = \frac{1}{z^{n+1}}$ has pole at $z = 0$ of order $(n+1)$

since pole is at $z = 0$ and is not on contour $|z| = 1$. we can integrate again by



$$z = e^{i\theta}$$

$$dz = e^{i\theta} i d\theta$$

$$\begin{aligned}
I &= \oint \frac{dz}{z^{n+1}} = \int_0^{2\pi} \frac{e^{i\theta} i d\theta}{e^{(n+1)i\theta}} \\
I &= \int_0^{2\pi} e^{-in\theta} i d\theta = \int_0^{2\pi} i (\cos n\theta - i \sin n\theta) d\theta \\
&= i \left(\frac{\sin n\theta}{n} + \frac{i \cos n\theta}{n} \right)_0^{2\pi} = 0. \quad \forall n = 1, 2, 3 \dots \\
&\quad (n \neq 0)
\end{aligned}$$

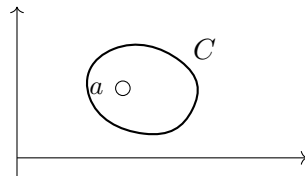
what if ($n = 0$),

$$I = \int_0^{2\pi} i e^{-in\theta} d\theta \Big|_{(n=0)} = \int_0^{2\pi} i d\theta = 2\pi i$$

So,

$$\boxed{\frac{1}{2\pi i} \oint \frac{dz}{z^{n+1}} = \delta_{n,0}}$$

we can generalize it,



$$\left\{ \frac{1}{2\pi i} \oint_C \frac{dz}{(z-a)^{n+1}} = \delta_{n,0} \right\}$$

So, If we have a function $f(z)$, having pole at $z = a$.

Then

$$\oint f(z) dz = ?$$

(or essential singularity)

As $f(z)$ has pole, we expand $f(z)$ about $z = a$ via laurent series.

$$f(z) = \sum_{n=1}^{\infty} \frac{c_{-n}}{(z-a)^n} + \sum_{n=0}^{\infty} c_n (z-a)^n$$

$$I = \oint f(z) dz = \sum_{n=1}^{\infty} \oint \frac{dz c_{-n}}{(z-a)^n} + \sum_{n=0}^{\infty} \oint \frac{dz c_n (z-a)^n}{(z-a)^n}$$

$\hookrightarrow$ is analytic function so vanishes

$$= (2\pi i) c_{-1} + \sum_{n=2}^{\infty} \oint \frac{dz c_{-n}}{(z-a)^n} + 0$$

$$\swarrow \text{ we just saw } \frac{1}{2\pi i} \oint \frac{dz}{(z-a)^n} = \delta_{n,0}$$

$$= (2\pi i) c_{-1} + 0$$

So,

$$\left\{ \frac{1}{2\pi i} \oint f(z) dz = \operatorname{Res}_{z=a} [f(z)] = c_{-1} \right\} \quad \begin{array}{l} \text{Cauchy} \\ \text{Integral} \\ \text{Formula} \end{array}$$

what if we find integral $\oint_C f(z) dz = ?$

here we are calculating integral over angle 0 to 4π i.e. 2 windings of contour.

So

$$\oint f(z) dz = \oint \sum_{n=1}^{\infty} \frac{c_{-n}}{(z-a)^n} + \sum_{n=0}^{\infty} \oint c_n (z-a)^n$$

$\hookrightarrow$ zero as $(z-a)^n$ is analytic.

$$\begin{aligned} &= (c_{-n}) \sum \int_0^{4\pi} \frac{dz}{(z-a)^{n+1}} + 0 \\ &= c_{-1} \int_0^{4\pi} e^{-in\theta} i d\theta + 0 \end{aligned}$$

✓ '0' (written over the n in $-in\theta$)

$$\oint f(z) dz = (4\pi i) \operatorname{Res}(z=a)$$

similarly for ' n ' consecutive loops.

$$\begin{aligned} \oint_{(n \times c)} f(z) dz &= n(2\pi i) \operatorname{Res}(z=a) \\ (c \div |z| &= 1) \end{aligned}$$

If we find integral in clockwise sense

$$\begin{aligned} \oint f(z) dz &= \sum_{n=1}^{\infty} \oint \frac{c_{-n}}{(z-a)^n} dz + \underbrace{\sum_{n=0}^{\infty} \oint c_n (z-a)^n}_{\text{Regular part so integral}=0} \\ &= \oint_c \frac{c_{-1} dz}{(z-a)} + \oint \sum_{n=2}^{\infty} \frac{c_{-n} dz}{(z-a)^n} + 0 \\ &= c_{-1} \oint \frac{dz}{(z-a)} + \sum_{n=2}^{\infty} \oint \frac{c_{-n} dz}{(z-a)^n} \\ &\quad \begin{cases} z-a = e^{i\theta} \\ dz = e^{i\theta} i d\theta \end{cases} \\ &= c_{-1} \int_{2\pi}^0 \frac{e^{i\theta}}{e^{i\theta}} i d\theta + \sum_{n=2}^{\infty} \oint_{2\pi}^0 \frac{c_{-n} e^{i\theta} i d\theta}{e^{in\theta}} \\ &= c_{-1} (-2\pi i) + \sum_{n=2}^{\infty} \left(\int_{2\pi}^0 c_{-n} e^{i\theta(1-n)} i d\theta \right) \end{aligned}$$

$$\rightarrow 0 \quad \forall n = 2, 3, 4 \dots$$

$$\oint f(z) dz = -2\pi i \operatorname{Res}(z = a)$$

In general,

$$\oint f(z) dz = n(2\pi i) \operatorname{Res}(z = a)$$

↓ no. of loops $(n \in \text{all Integers})$

example.

$$(1) \quad I = \oint e^{1/z} dz$$

$$\begin{aligned} I &= \oint \left(1 + \frac{1}{z} + \frac{1}{2!z^2} + \dots \right) dz \\ &= \oint \frac{\boxed{1}}{z} dz + 0 \quad (\operatorname{Res}(z = 0) = 1) \\ &\quad (I = 2\pi i). \end{aligned}$$

$$(2) \quad \oint e^{\frac{1}{z^2}} dz = 2\pi i \operatorname{Res}(f(z)) = 0$$

$$e^{\frac{1}{z^2}} = 1 + \frac{1}{z^2} + \frac{1}{2!z^4} + \dots$$

$$\operatorname{Res} = \left(\text{coeff of } \frac{1}{(z-0)} = \boxed{0} \right)$$